

DSA0503 – Query Processing for Data Science

LAB EXPERIMENTS

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Experiment 1

Aim

To select the distinct Department IDs from the employees dataset using Pandas.

Algorithm

Import the Pandas library.

Read the employees dataset.

Select the Department_ID column.

Find the distinct department IDs.

Display the distinct department IDs.

Output

```
Distinct Department IDs:
0      10
1      20
2      30
3      40
4      50
5      60
6      70
7      80
8      90
9     100
10     110
11     120
12     130
13     140
14     150
15     160
16     170
17     180
18     190
19     200
20     210
21     220
22     230
23     240
24     250
25     260
26     270
Name: DEPARTMENT_ID, dtype: int64
```

Result

The distinct Department IDs were successfully extracted and displayed.

Experiment 2

Aim

To display the IDs of employees who have done two or more jobs in the past using Pandas.

Algorithm

Import the Pandas library.

Read the employee job history dataset.

Group the records based on Employee ID.

Count the number of jobs performed by each employee.

Display the Employee IDs having two or more job records.

Output

Employee Job History:

	EMPLOYEE_ID	START_DATE	END_DATE	JOB_ID	DEPARTMENT_ID
0	102	2001-01-13	2006-07-24	IT_PROG	60
1	101	1997-09-21	2001-10-27	AC_ACCOUNT	110
2	101	2001-10-28	2005-03-15	AC_MGR	110
3	201	2004-02-17	2007-12-19	MK_REP	20
4	114	2006-03-24	2007-12-31	ST_CLERK	50
5	122	2007-01-01	2007-12-31	ST_CLERK	50
6	200	1995-09-17	2001-06-17	AD_ASST	90
7	176	2006-03-24	2006-12-31	SA_REP	80
8	176	2007-01-01	2007-12-31	SA_MAN	80
9	200	2002-07-01	2006-12-31	AC_ACCOUNT	90

Employees who have done two or more jobs:

	EMPLOYEE_ID
0	101
4	176
5	200

Result

The Employee IDs with two or more job records were successfully displayed.

Experiment 3

Aim

To display the details of jobs in descending order based on the Job Title using Pandas.

Algorithm

Import the Pandas library.

Read the jobs dataset.

Sort the records in descending order based on the Job Title.

Display the sorted dataset.

Output

Job Details in Descending Order of Job Title:

	JOB_ID	JOB_TITLE	MIN_SALARY	MAX_SALARY
11	ST_MAN	Stock Manager	5500	8500
12	ST_CLERK	Stock Clerk	2008	5000
13	SH_CLERK	Shipping Clerk	2500	5500
8	SA_REP	Sales Representative	6000	12008
7	SA_MAN	Sales Manager	10000	20080
9	PU_MAN	Purchasing Manager	8000	15000
10	PU_CLERK	Purchasing Clerk	2500	5500
18	PR_REP	Public Relations Representative	4500	10500
6	AC_ACCOUNT	Public Accountant	4200	9000
14	IT_PROG	Programmer	4000	10000
0	AD PRES	President	20080	40000
16	MK_REP	Marketing Representative	4000	9000
15	MK_MAN	Marketing Manager	9000	15000
17	HR_REP	Human Resources Representative	4000	9000
3	FI_MGR	Finance Manager	8200	16000
1	AD_VP	Administration Vice President	15000	30000
2	AD_ASST	Administration Assistant	3000	6000
5	AC_MGR	Accounting Manager	8200	16000
4	FI_ACCOUNT	Accountant	4200	9000

Result

The job details were successfully sorted and displayed in descending order.

Experiment 4

Aim

To create a line plot of the historical stock prices of Alphabet Inc. between two specific dates using Pandas and Matplotlib.

Algorithm

Import the required libraries.

Read the Alphabet stock dataset.

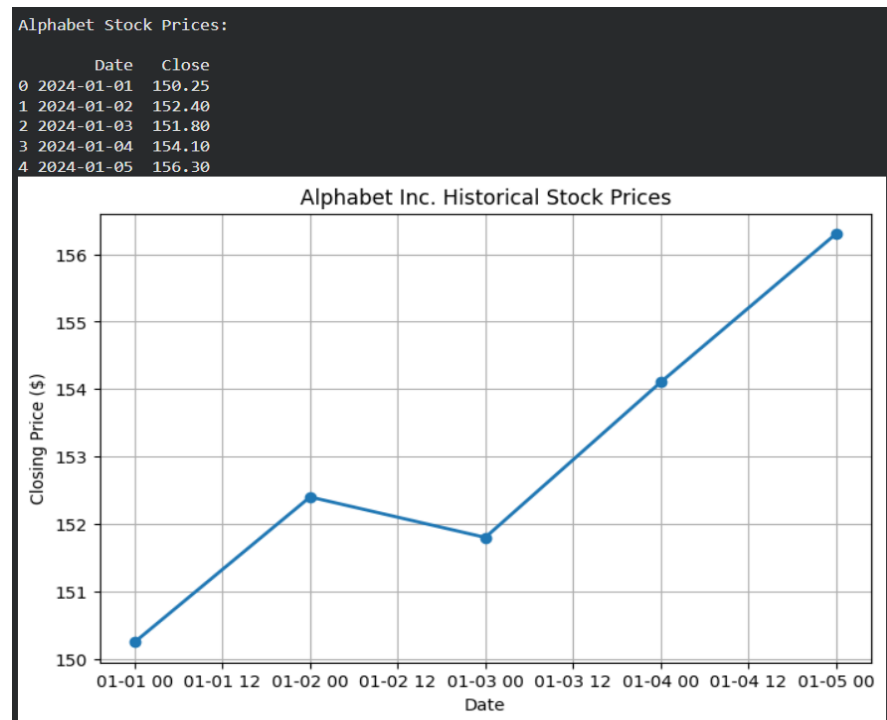
Convert the Date column into date format.

Filter the records between the specified dates.

Plot the stock prices using a line graph.

Display the graph.

Output



Result

The line plot of the historical stock prices was generated successfully.

Experiment 5

Aim

To create a bar plot of the trading volume of Alphabet Inc. stock between two specific dates.

Algorithm

Import the required libraries.

Read the Alphabet stock dataset.

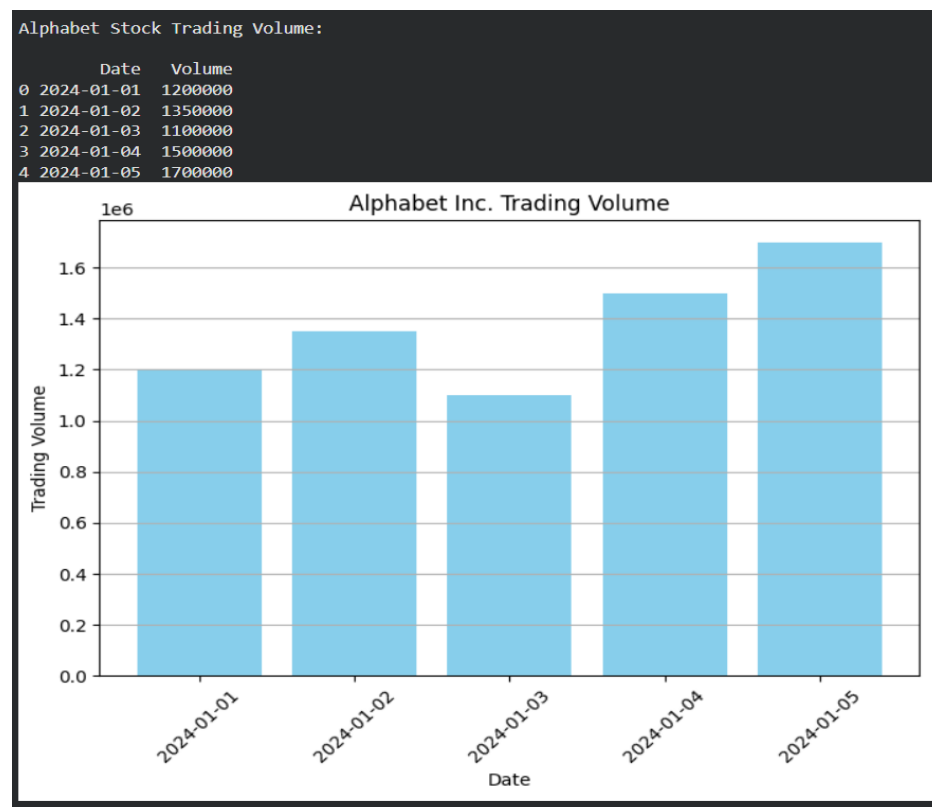
Convert the Date column into date format.

Filter the records between the specified dates.

Create a bar chart of the trading volume.

Display the chart.

Output



Result

The bar plot of the trading volume was generated successfully.

Experiment 6

Aim

To create a scatter plot of the trading volume and stock prices of Alphabet Inc. between two specific dates.

Algorithm

Import the required libraries.

Read the Alphabet stock dataset.

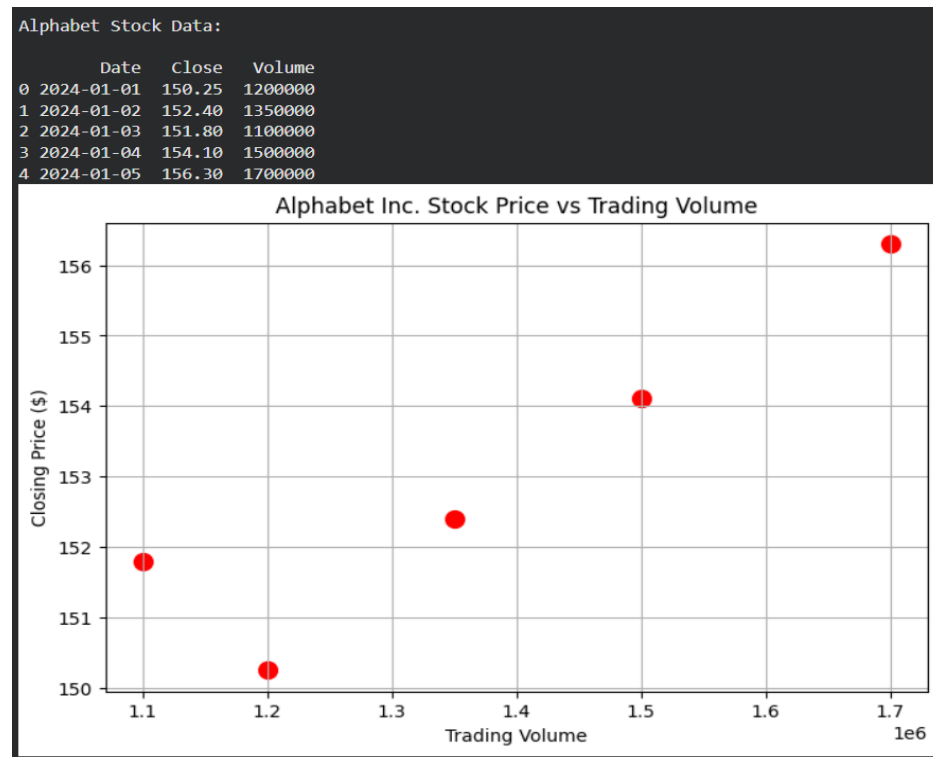
Convert the Date column into date format.

Filter the required records.

Create a scatter plot between trading volume and stock prices.

Display the graph.

Output



Result

The scatter plot was generated successfully.

Experiment 7

Aim

To create a Pivot Table and find the maximum and minimum sale value of the items using the sales dataset.

Algorithm

Import the Pandas library.

Read the sales dataset.

Create a Pivot Table using the Item column.

Calculate the maximum and minimum sale values.

Display the Pivot Table.

Output

```
Maximum and Minimum Sale Amount of Each Item:
```

	max	min
Item	Sale_amt	Sale_amt
Cell Phone	14400	6075
Desk	250	250
Home Theater	40500	14000
Television	113810	38336
Video Games	936	936

Result

The Pivot Table was successfully created and the maximum and minimum sale values were obtained.

Experiment 8

Aim

To create a Pivot Table and find the item-wise units sold using the sales dataset.

Algorithm

Import the Pandas library.

Read the sales dataset.

Create a Pivot Table using the Item column.

Calculate the total units sold for each item.

Display the Pivot Table.

Output

Item-wise Total Units Sold:

Item	Units
Cell Phone	91
Desk	2
Home Theater	308
Television	509
Video Games	16

Result

The Pivot Table was successfully created and the item-wise units sold were displayed.

Experiment 9

Aim

To create a Pivot Table and find the total sale amount region-wise, manager-wise, and salesman-wise using the sales dataset.

Algorithm

Import the Pandas library.

Read the sales dataset.

Create a Pivot Table using Region, Manager, and Salesman.

Calculate the total sale amount.

Display the Pivot Table.

Output

Total Sale Amount (Region-wise, Manager-wise, Salesman-wise):

			Sale_amt
Region	Manager	SalesMan	
Central	Douglas	John	250
		Hermann	150948
		Shelli	25000
		Sigal	121820
	Martha	Steven	89850
	Timothy	David	6075
East	Douglas	Karen	40500
	Martha	Alexander	231076
		Diana	14500
West	Douglas	Michael	38336
	Timothy	Stephen	67088

Result

The Pivot Table was successfully created and the total sale amount was displayed.

Experiment 10

Aim

To create a DataFrame with random values and highlight negative numbers in red and positive numbers in black using Pandas styling.

Algorithm

Import the required libraries.

Create a DataFrame with random values.

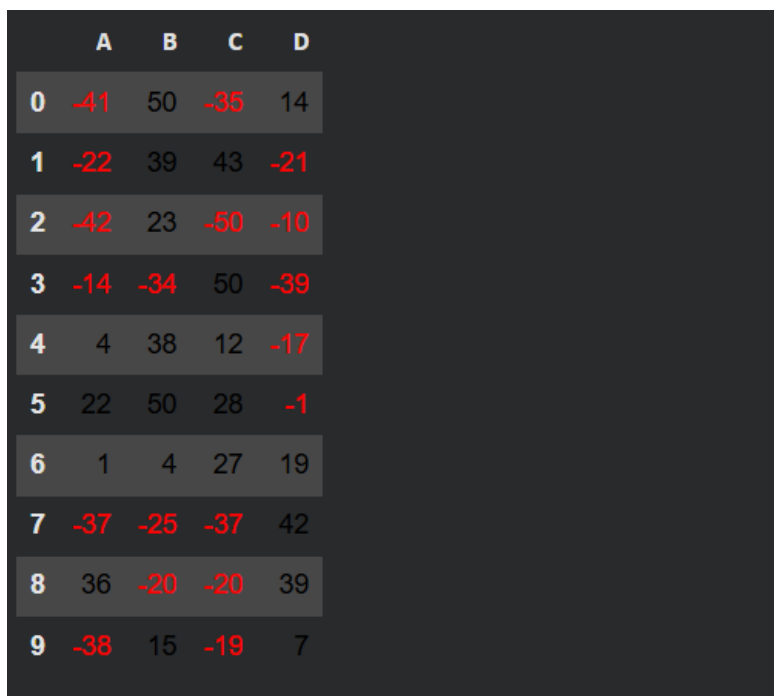
Apply conditional formatting to the DataFrame.

Highlight negative values in red.

Highlight positive values in black.

Display the styled DataFrame.

Output



	A	B	C	D
0	-41	50	-35	14
1	-22	39	43	-21
2	-42	23	-50	-10
3	-14	-34	50	-39
4	4	38	12	-17
5	22	50	28	-1
6	1	4	27	19
7	-37	-25	-37	42
8	36	-20	-20	39
9	-38	15	-19	7

Result

The DataFrame was successfully created and the values were highlighted based on their sign.